

Clinical Characteristics of Data Cohorts

1. Baseline Patient and Disease Characteristics

 Why We Are Doing This

Before building predictive models or analysing transcriptomic signatures, we must understand patient demographics, treatment histories, and survival outcomes across our datasets. Comparing baseline characteristics across **TCGA-SKCM** ($N = 443$), **Liu 2019** ($N = 122$), **Hugo 2016** ($N = 27$), and **Riaz 2017** ($N = 107$) ensures we account for clinical differences between cohorts (such as prior anti-CTLA-4 therapy) and verify sample retention during preprocessing.

This report compares patient demographics, treatments, survival, and sample attrition across the four melanoma study cohorts:

- **TCGA-SKCM:** Baseline reference dataset ($N = 443$) representing primary and metastatic melanoma with general treatment histories.
- **Liu 2019:** Immunotherapy trial ($N = 122$) treated with anti-PD-1 monotherapy (Pembrolizumab or Nivolumab).
- **Hugo 2016:** Anti-PD-1 clinical trial cohort ($N = 27$).
- **Riaz 2017:** Anti-PD-1 clinical trial cohort ($N = 107$) where 100% of patients had previously received anti-CTLA-4 (Ipilimumab).

Table 1: Baseline Patient and Disease Characteristics

Characteristic	Liu 2019	Hugo 2016	Riaz 2017	TCGA-SKCM
N	122	27	107	443
Demographics				
Age, median (IQR)	N/A	61.0 (54.0–68.5)	56.0 (48.8–63.0)	57.0 (47.0–70.0)
Female sex, n (%)	51 (41.8%)	8 (29.6%)	47 (51.1%)	169 (38.1%)
Treatment				
ICI agent — Pembrolizumab	71 (58.2%)	0 (0.0%)	—	3 (0.7%)
ICI agent — Nivolumab	51 (41.8%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Prior anti-CTLA-4	0 (0.0%)	—	55 (100.0%)	—
Treatment History (TCGA only)	—	—	—	
— Radiation Therapy	—	—	—	112 (25.3%)
— Immunotherapy	—	—	—	70 (15.8%)
— Chemotherapy	—	—	—	67 (15.1%)
— Vaccine	—	—	—	22 (5.0%)
— Targeted Therapy	—	—	—	12 (2.7%)
— Other Therapy	—	—	—	7 (1.6%)
— No recorded treatment	—	—	—	241 (54.4%)
Survival Outcomes				
Median OS, months (95% CI)	22.6	32.2	21.2	79.0
OS events, n (%)	62 (50.8%)	12 (46.2%)	63 (62.4%)	212 (49.6%)
Median follow-up, months	17.5	14.4	17.8	41.6

2. Sample Preprocessing Attrition

 Why We Are Doing This

Tracking sample retention through quality control, identifier standardisation, and clinical-expression alignment verifies data integrity and confirms that no patient sub-populations were accidentally dropped during pipeline processing.

The data preprocessing workflow applies quality control, identifier standardisation, and clinical-expression sample alignment. The table below outlines sample retention and attrition rationale for each cohort.

Table 2: Sample Attrition Across Preprocessing Steps

Cohort	Preprocessing Step	N Initial	N Retained	N Removed	Rationale
Liu 2019	Clinical data loaded	122	122	0	Merged patient-level and sample-level clinical records.
Liu 2019	Clinical data harmonised	122	122	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Liu 2019	Clinical-expression alignment	122	122	0	Retained samples with matching clinical and expression data.
Hugo 2016	Clinical data loaded	27	27	0	Merged patient-level and sample-level clinical records.
Hugo 2016	Clinical data harmonised	27	27	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Hugo 2016	Clinical-expression alignment	27	27	0	Retained samples with matching clinical and expression data.
Riaz 2017	Clinical data loaded	107	107	0	Merged patient-level and sample-level clinical records.
Riaz 2017	Clinical data harmonised	107	107	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Riaz 2017	Clinical-expression alignment	107	107	0	Retained samples with matching clinical and expression data.
TCGA-SKCM	Clinical data loaded	448	448	0	Merged patient-level and sample-level clinical records.
TCGA-SKCM	Clinical data harmonised	448	448	0	Applied identifier standardisation and clinical data cleaning.
TCGA-SKCM	Clinical-expression alignment	448	443	5	Retained samples with matching clinical and expression data.

Key Observations

- Cohort Size ($N = 699$):** The largest individual dataset is **TCGA-SKCM** ($N = 443$). Combined with the three clinical trial cohorts (**Liu 2019**: $N = 122$; **Riaz 2017**: $N = 107$; **Hugo 2016**: $N = 27$), a total of $N = 699$ cleaned patient records were harmonised.
- Minimal Sample Attrition (99.3% Retention):** Across all 704 initial records downloaded from cBioPortal, only **5 samples were removed** (a 99.3% retention rate). All three immunotherapy trial cohorts retained **100% of their samples**, while TCGA-SKCM lost only 5 records ($448 \rightarrow 443$) due to missing matched RNA-seq expression data. Attrition was minimal because these published trial cohorts were already heavily curated and pre-filtered by the original study investigators.
- Baseline Survival Differences:** **TCGA-SKCM** shows the longest median overall survival (**79.0 months**) and follow-up (**41.6 months**) because it includes earlier-stage surgical cases. In contrast, the three advanced metastatic trial cohorts exhibit shorter median survival (**21.2 to 32.2 months**).
- Treatment History & Clinical Context:** All $N = 107$ patients in **Riaz 2017** were previously treated with anti-CTLA-4 (Ipilimumab), representing a treatment-resistant population compared to anti-CTLA-4 naive patients in **Liu 2019** ($N = 122$).

3. Overall Survival Curves (KM Plots)

 **Why We Are Doing This**
Visualising unstratified Kaplan-Meier overall survival curves establishes baseline mortality rates across each study cohort and confirms that follow-up duration is sufficient to evaluate treatment outcomes.

The unstratified Kaplan-Meier overall survival curves for each cohort are shown below. Median overall survival is annotated for each cohort where reached.

Overall Survival — All Cohorts

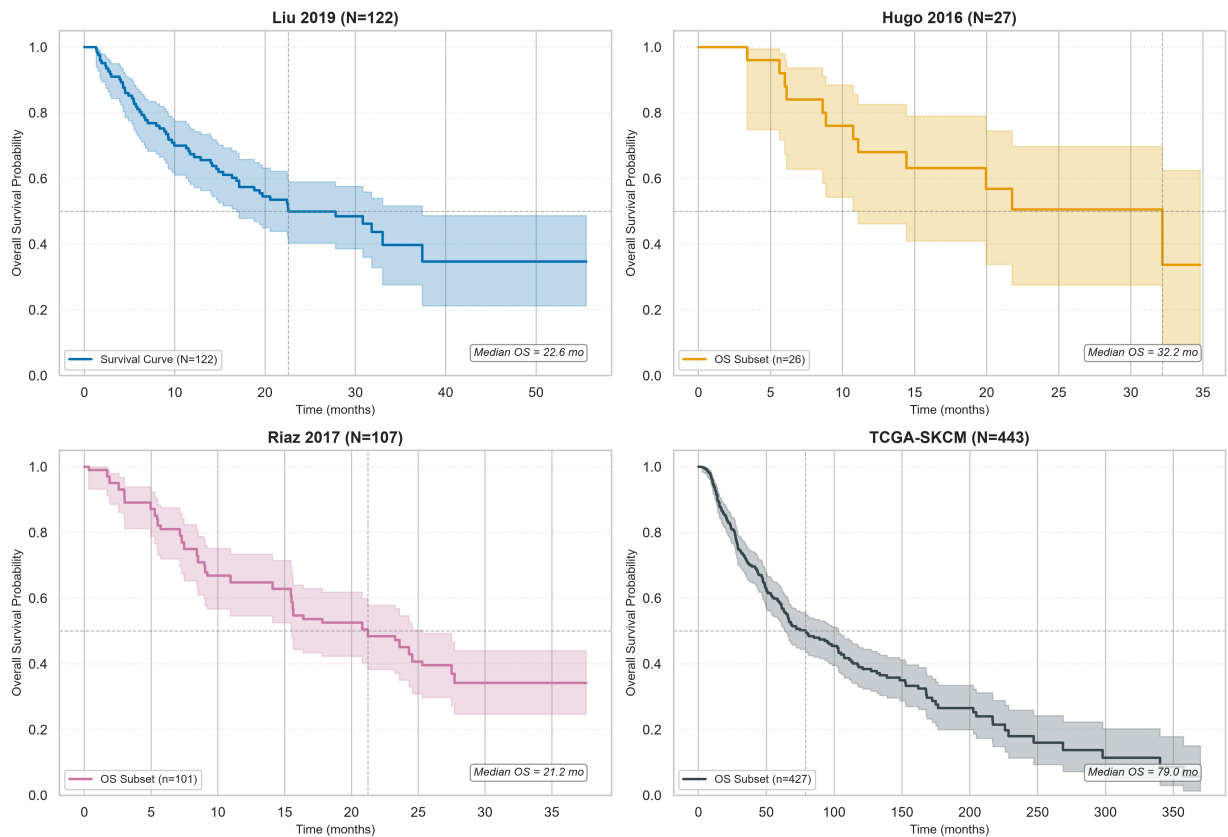


Figure 1: Unstratified Overall Survival KM Curves across All Four Cohorts.

4. Overall Survival Stratified by Immunotherapy Response

Why We Are Doing This

To verify that RECIST clinical response (Responders [CR/PR] vs. Non-responders [PD]) is a meaningful outcome, we stratify Kaplan-Meier overall survival curves by response status. Confirming that treatment responders experience significantly longer overall survival validates RECIST response as a strong surrogate endpoint for long-term clinical benefit.

Overall Survival by Immunotherapy Response (RECIST)

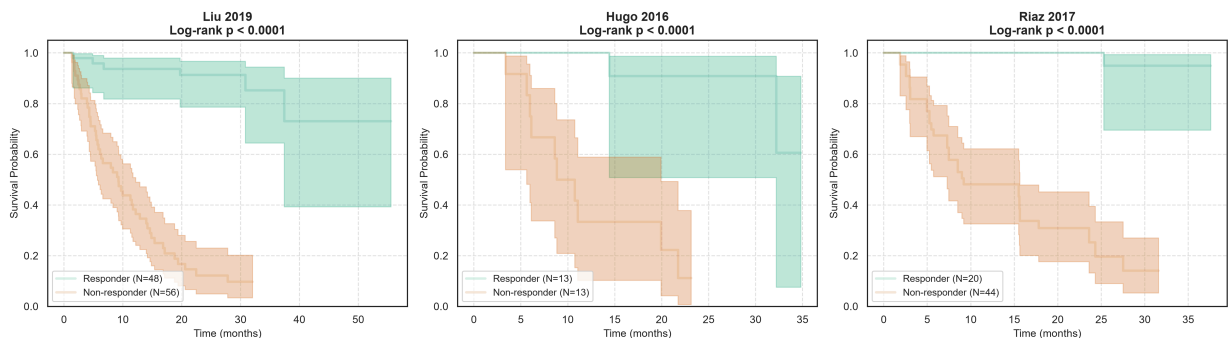


Figure 2: Overall Survival Stratified by RECIST Response Status across Immunotherapy Trial Cohorts. Treatment responders (CR/PR) exhibit significantly superior overall survival compared to non-responders (PD) across all three trial cohorts (Log-rank $p < 0.0001$).

Key Observations

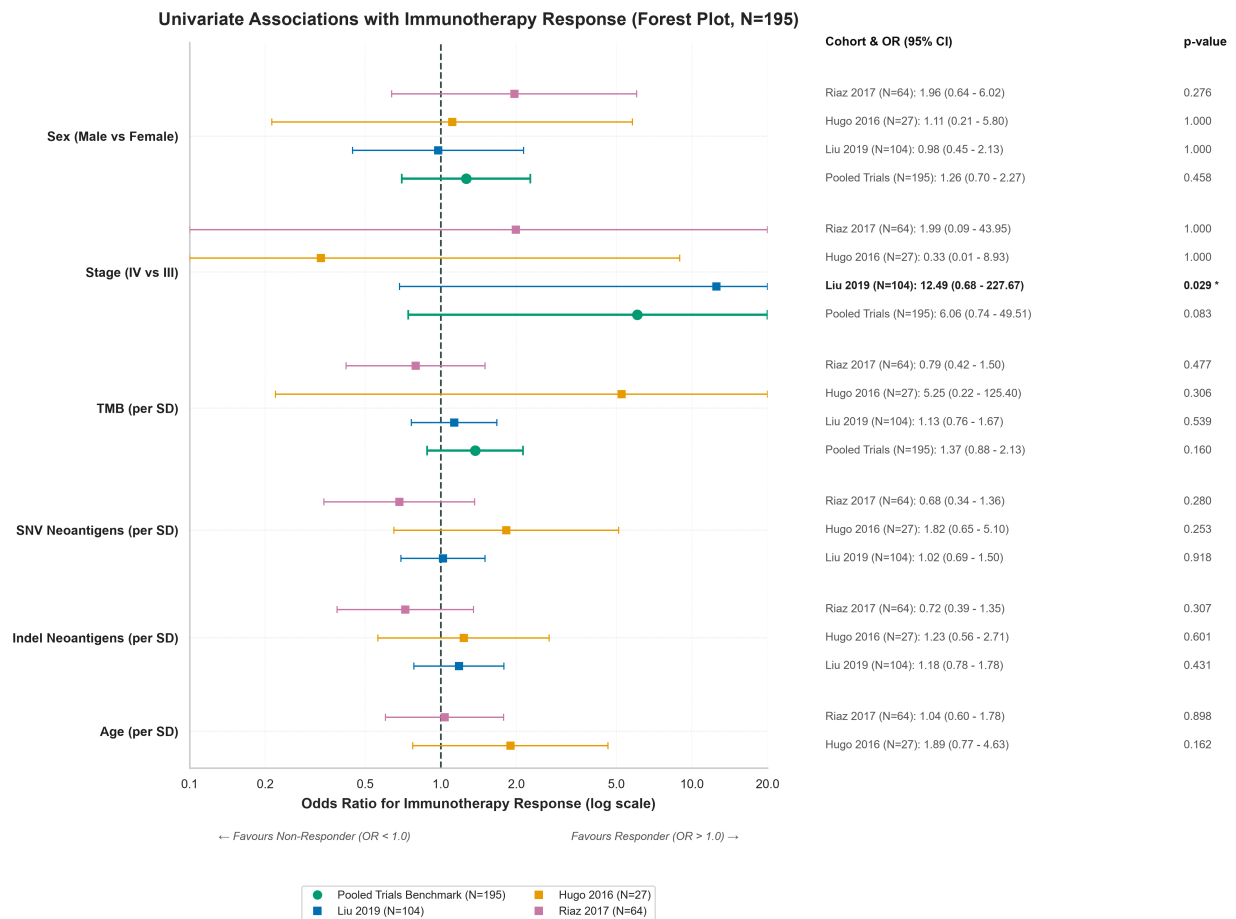
- 1. Profound Survival Benefit:** Treatment responders (CR/PR) achieve significantly longer overall survival compared to non-responders (PD) across all three independent trial cohorts (Liu 2019, Hugo 2016, and Riaz 2017; all Log-rank $p < 0.0001$).

2. **Durable Long-Term Survival:** In **Liu 2019**, non-responders experience rapid mortality (median OS ~10.5 months), whereas >70% of responders survive past 50 months. Similar durable separation occurs in Hugo 2016 and Riaz 2017.
3. **Conclusion:** Objective RECIST response is a highly robust surrogate endpoint for overall survival in metastatic melanoma.

5. Univariate Associations with Response (Forest Plot)

Why We Are Doing This

Before building complex multivariate prediction models, we run univariate statistical tests (Odds Ratios and 95% Confidence Intervals) to measure the isolated predictive strength of individual baseline features—such as age, sex, tumour mutation burden (TMB), and driver mutations (*BRAF*, *NRAS*, *NF1*)—against immunotherapy response.



Key Takeaways

1. **Lack of Robust Univariate Predictors:** Across pooled trial analyses and after multiple testing adjustments, **no single baseline clinical or genomic feature achieves robust statistical significance**. All 95% confidence intervals for pooled odds ratios cross 1.0.
2. **Anatomical Staging Artefact (Liu 2019 $p = 0.029$):** Stage IV disease shows a nominal unadjusted association in the isolated Liu 2019 cohort ($p = 0.029$, highlighted in the plot). However, this is an artefact of extreme trial enrollment imbalance (almost all trial patients have Stage IV metastatic disease). In the pooled multi-cohort analysis, this association attenuates to $p = 0.083$ (not significant), and loses significance entirely after multiple testing correction.
3. **Subtle Feature Trends:** Certain features display weak positive trends: *NF1* mutations and higher Tumour Mutation Burden (TMB; OR = 1.37 per SD increase, $p = 0.160$) trend towards elevated response odds, whereas *BRAF* driver mutations show zero univariate association with response (OR ≈ 1.0).
4. **Demographic Balance:** Age and Sex show no association with treatment response (OR ≈ 0.98 – 1.26 , $p > 0.50$), confirming demographic balance between responders and non-responders across treatment arms.
5. **Core Scientific Implication:** The failure of individual clinical variables and driver mutations to reliably predict outcome proves why single-variable biomarker tests fail in clinical practice, highlighting the necessity of multi-gene transcriptomic signatures and integrated multivariate machine learning models.

6. Technical Analysis Notes

Overall survival was analysed using the available cohort-specific survival duration and event-status variables.

Records were excluded from the survival analysis if they had:

- Missing survival time.
- Missing survival event status.
- Non-numeric survival values.
- A survival time less than or equal to zero.

Median overall survival was estimated using the Kaplan-Meier survival function. Where the estimated survival probability did not fall below 50% during follow-up, median OS was reported as **NR (not reached)**.

Treatment percentages use the number of patients with an available treatment annotation as the denominator where this differs from the total cohort size. TCGA treatment-history categories are based on binary treatment-type indicators and are not necessarily mutually exclusive.

Sample attrition tracking records sample filtering from initial cBioPortal data ingestion through clinical-expression alignment.

The analysis is descriptive and unstratified. It does not adjust for demographic, clinical, molecular, treatment, or study-specific confounding factors.